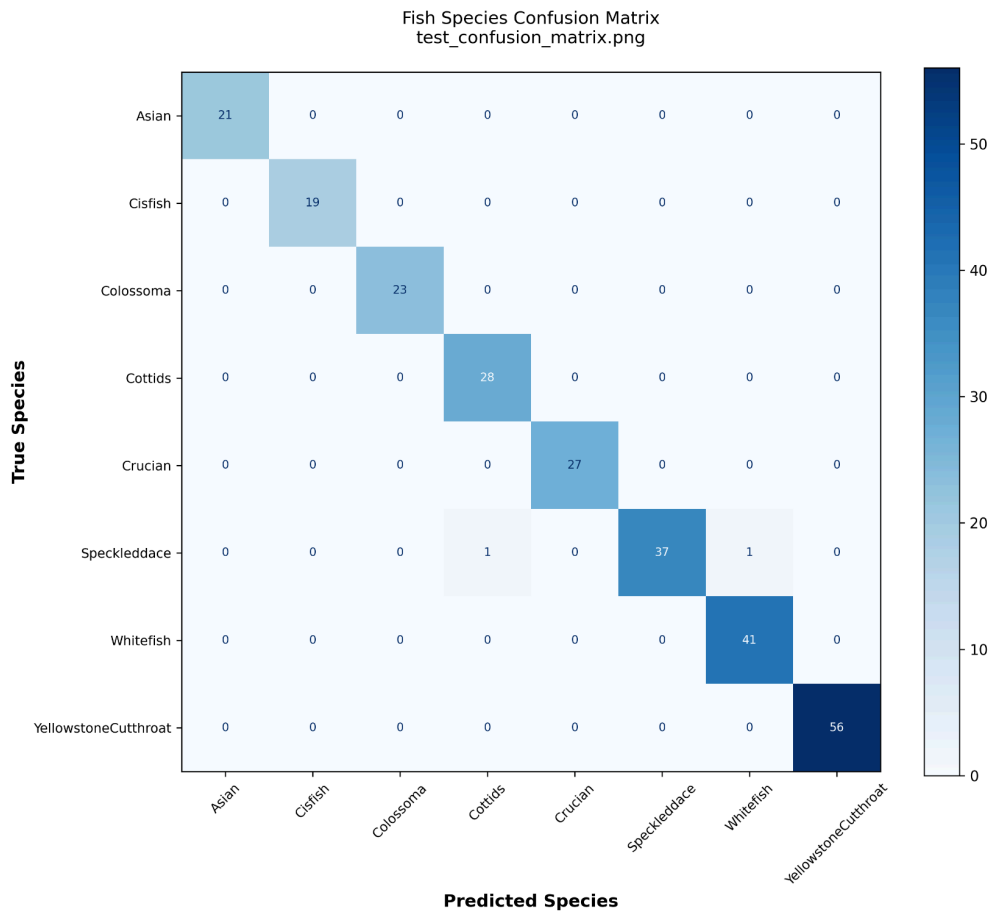


Classification Homework

Allan Howe - Robotic Vision ECEN 631

The Fish Dataset

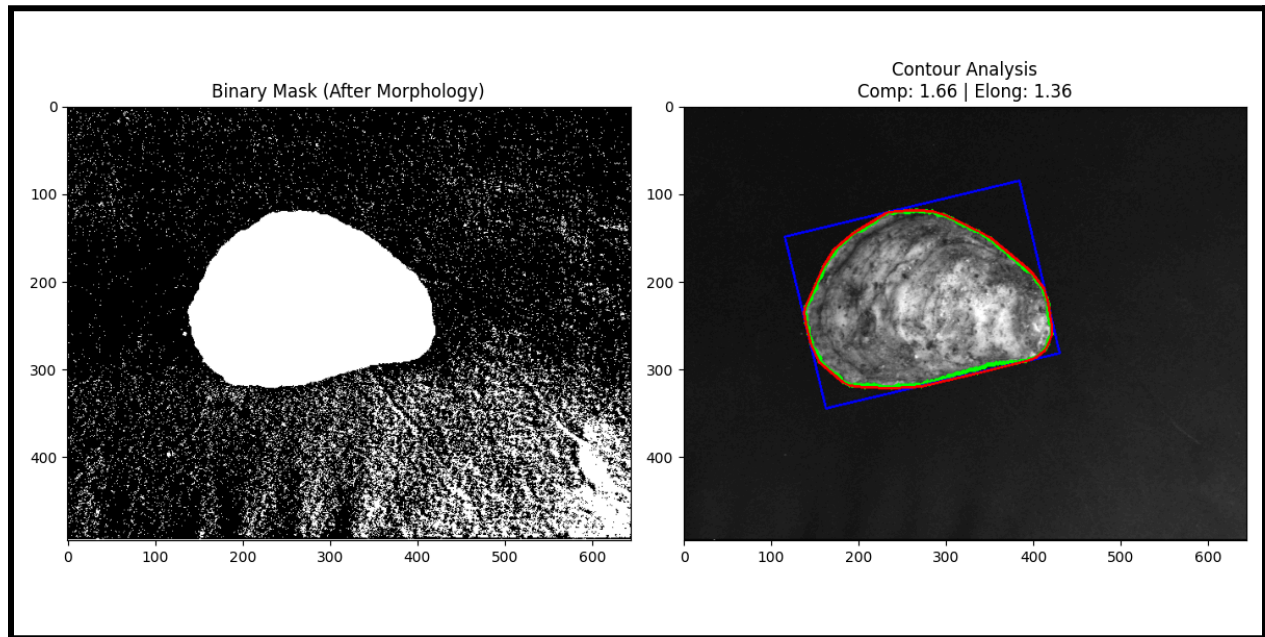
I used pytorch, and I used the [example MNIST](#) dataset as a template, and adjusted it for color images, as well as pooling size for my images. I was going to adjust it further, but it worked so well I did not need to make any more adjustments. I made the fish all 64x64 input images. I augmented the dataset just doing color jitter, and normalization. The model ended up being 99% accurate! Which is great performance.



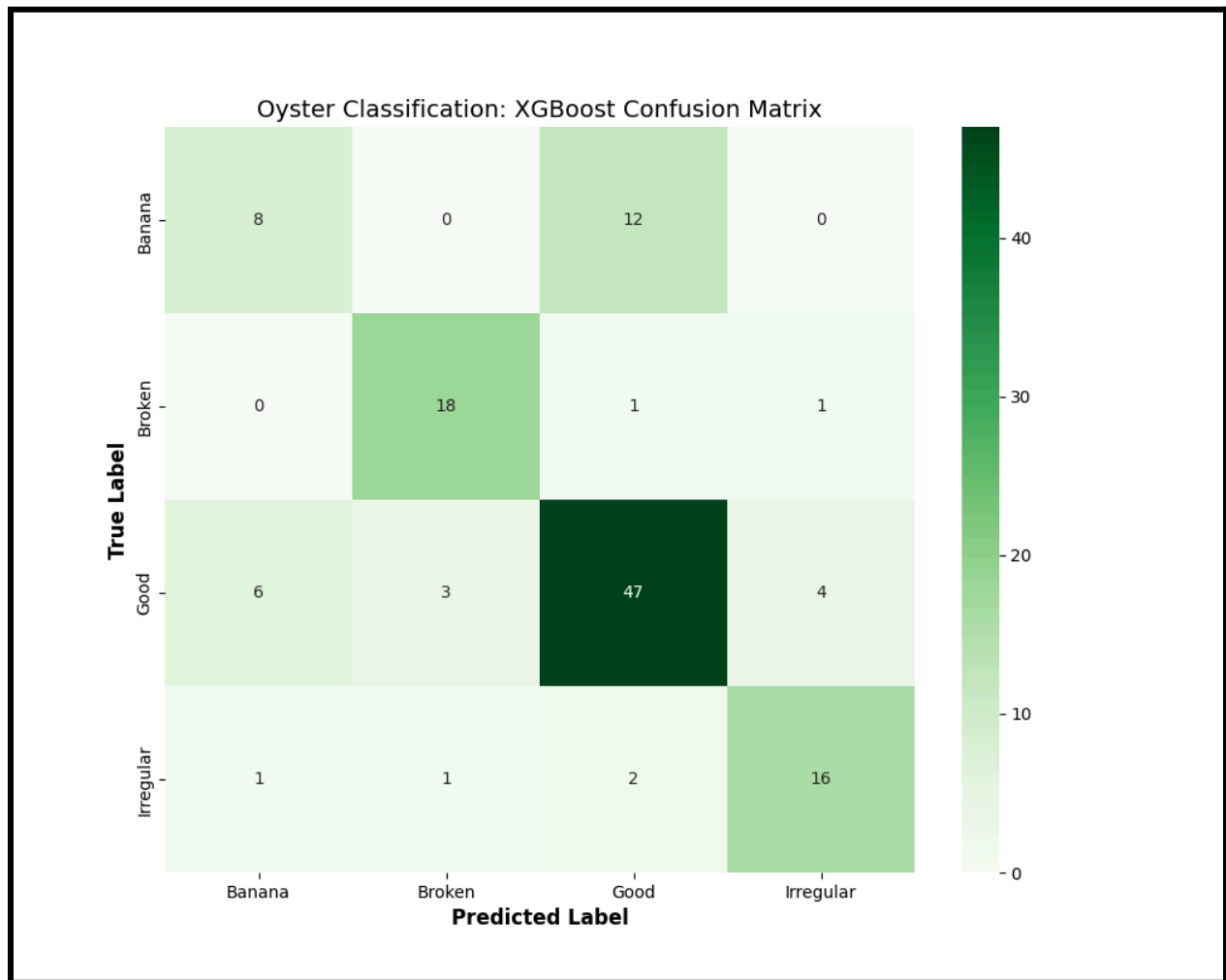
The Oyster Dataset

This dataset was a lot trickier. I first tried a CNN on it, the same one from the fish dataset. That did not do well at all. I tried doing some pre-modifications and removing the background and cropping, but that still did not prove very helpful. Eventually, I followed some smart papers that have done this before, and extracted features, and used a XGboost classification model. This finally got my model pretty accurate at 75%.

This images shows some of the features I extracted:



And the confusion matrix for this model was as follows:



I think one of the issues here was for the model to classify the good ones vs the banana shaped ones, sometimes the elongation factor may overlap. This was the best I could get the model to do after several hours of training. I think it is pretty good.